

Headline part VI: the equal-ratio normal moment integral

Claude, with Daniel Murfet

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Abstract

Paper-facing wrappers for the general- d equal-ratio monomial milestone (units 165, 170–175), stated directly on the Bochner box integral $M(N) = \int_{(0,1]^{m+1}} \prod_i x_i^{h_i} e^{-\beta N \prod_i x_i^{2k_i}} dx$ (the paper's bare normal moment integral with cutoff $b = 1$): the product density, the Gamma/log asymptotic, the exact reduction $M(N) = \prod_i \frac{1}{2k_i} \cdot \frac{1}{m!} \int_0^1 z^{\lambda-1} (-\log z)^m e^{-\beta N z} dz$, and the asymptotic $M(N) \sim \Gamma(\lambda) \beta^{-\lambda} / (m! \prod_i 2k_i) N^{-\lambda} (\log N)^m$ in ratio and paper forms. Zero sorry/axiom.

1 The things formalised

```
theorem headline_product_density (l) (m) (g) (hg : Measurable g) :
  weightedBoxIntegral (m + 1) (fun _ => 1 - 1) g
    = ofReal (1 / m.factorial) *   z in Ioc 0 1, ofReal (logDensity l m z) * g z
theorem headline_gamma_log_asymptotic (hl : 0 < l) (h : 0 < ) :
  (fun N =>   z in Ioc 0 1, z ^ (l - 1) * (-log z) ^ m * exp (-( * N * z)))
    ~[atTop] fun N => Gamma l *   ^ (-l) * N ^ (-l) * log N ^ m
theorem headline_normal_moment_reduction (m) (h k : Fin (m+1) → ) (hk) (l N) (hl)
  (hratio :   i, ((h i : ) + 1) / (2 * k i) = l) (hN : 0 < * N) :
  x in unitBox (m + 1), (   i, x i ^ h i) * exp (-( * N *   i, x i ^ (2 * k i)))
    = (   i, 1 / (2 * k i)) * (1 / m.factorial)
      *   z in Ioc 0 1, z ^ (l - 1) * (-log z) ^ m * exp (-( * N * z))
theorem headline_normal_moment_ratio ... : Tendsto (fun N => (   x in unitBox (m + 1), ...)
  / (N ^ (-l) * log N ^ m)) atTop
  ( ((   i, 1 / (2 * k i)) * (1 / m.factorial) * (Gamma l *   ^ (-l))))
theorem headline_normal_moment_asymptotic ... :
  (fun N =>   x in unitBox (m + 1),
    (   i, x i ^ h i) * exp (-( * N *   i, x i ^ (2 * k i))))
    ~[atTop] fun N => Gamma l *   ^ (-l) / (m.factorial *   i, 2 * k i)
      * N ^ (-l) * log N ^ m
```

2 Paper-facing meaning

The paper's zeta-function analysis of the bare normal moment integral (§“Standard form and the normal moment integrals”) predicts the leading exponent $\lambda = \min_i (h_i + 1) / (2k_i)$ with multiplicity $|J|$, J the set of coordinates realising the minimum, and the constant $\Gamma(\lambda) a_{-|J|} / (|J| - 1)!$ with $a_{-|J|} = \prod_{i \in J} \frac{1}{2k_i}$ when $b = 1$ and $J = I$. The wrappers formalise exactly this equal-ratio boundary case, for arbitrary $\beta > 0$, without any Taylor-tree or quadratic-chart intermediary. Not claimed: unequal ratios, cutoffs $b \neq 1$, interior coordinates with parity factors.

3 Lean notes

The wrappers unfold `monomialBoxReal`, `gammaLogIntegral` and `logDensity` definitionally, so each is a one-line `exact` of the unit-175 theorem; no `show/change` was needed because the definitions are reducible enough for `exact`-level unification of the integrands.